PRACTICAL WORK 3

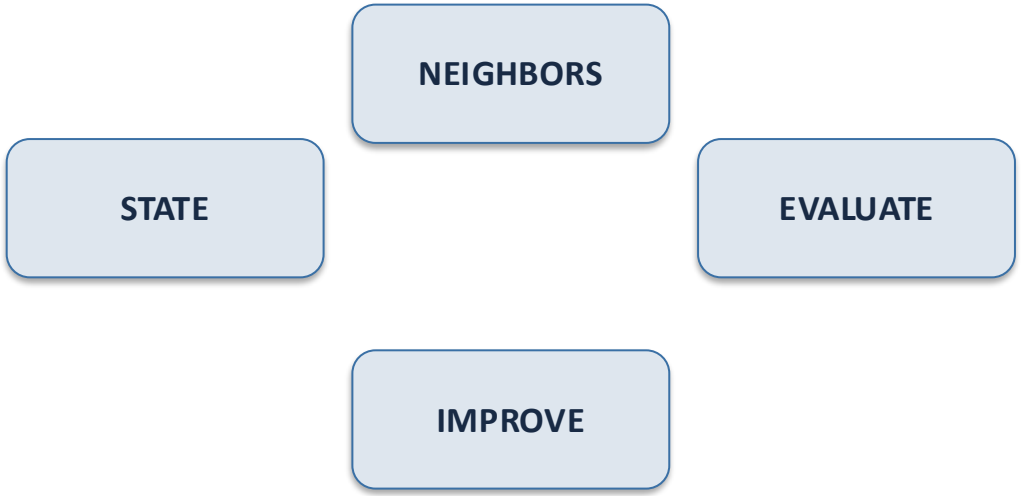
Local Search: Hill Climbing, Random Restart & Simulated Annealing

Goal: improve a complete current solution by exploring neighboring states.

1. Local Search in One Idea

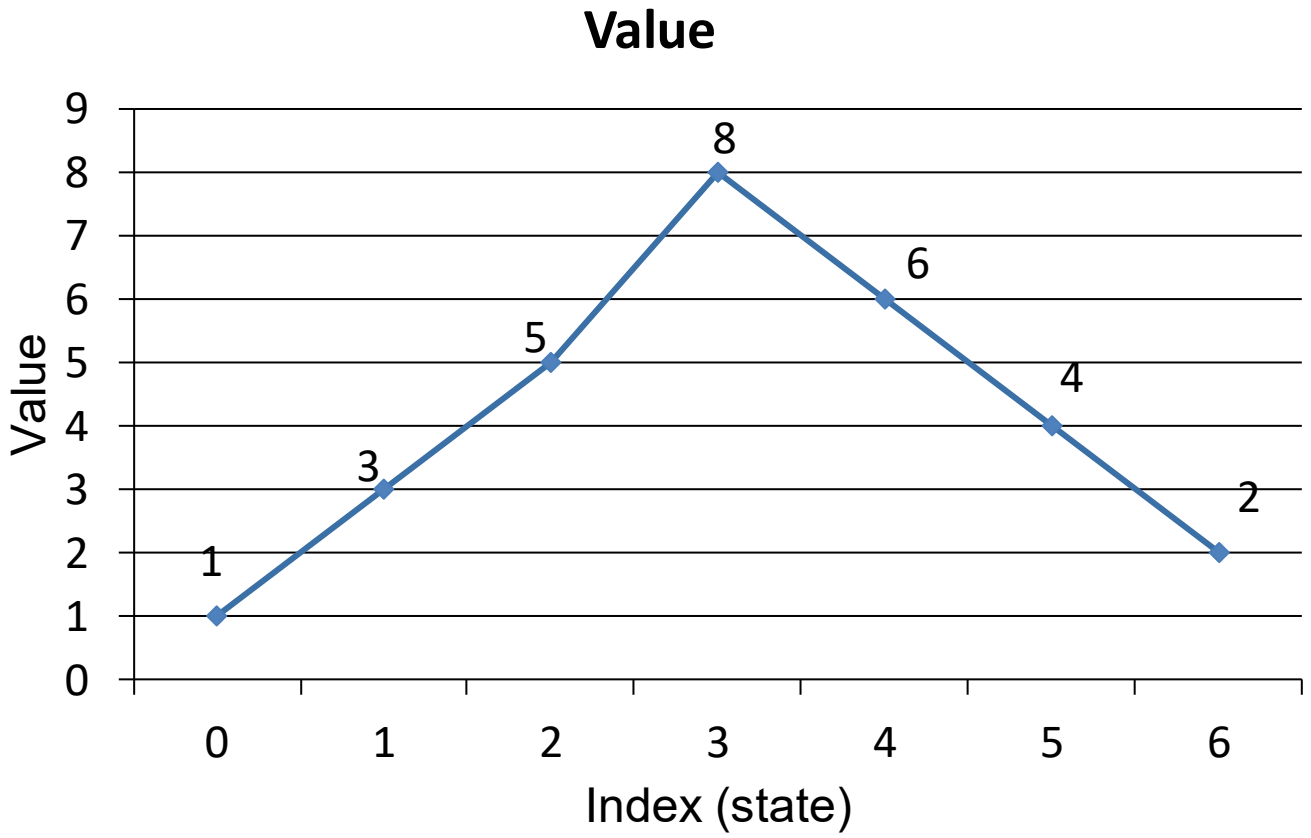
Unlike BFS/DFS/UCS/A*, the path is often not the solution.

- State = current complete candidate solution
- Neighbor = a state obtained by a small change
- Objective / value function = numerical quality of the state
- Main cycle: STATE → NEIGHBORS → EVALUATE → IMPROVE



2. Hill Climbing — Step-by-Step Example

values = [1, 3, 5, 8, 6, 4, 2], start = index 0

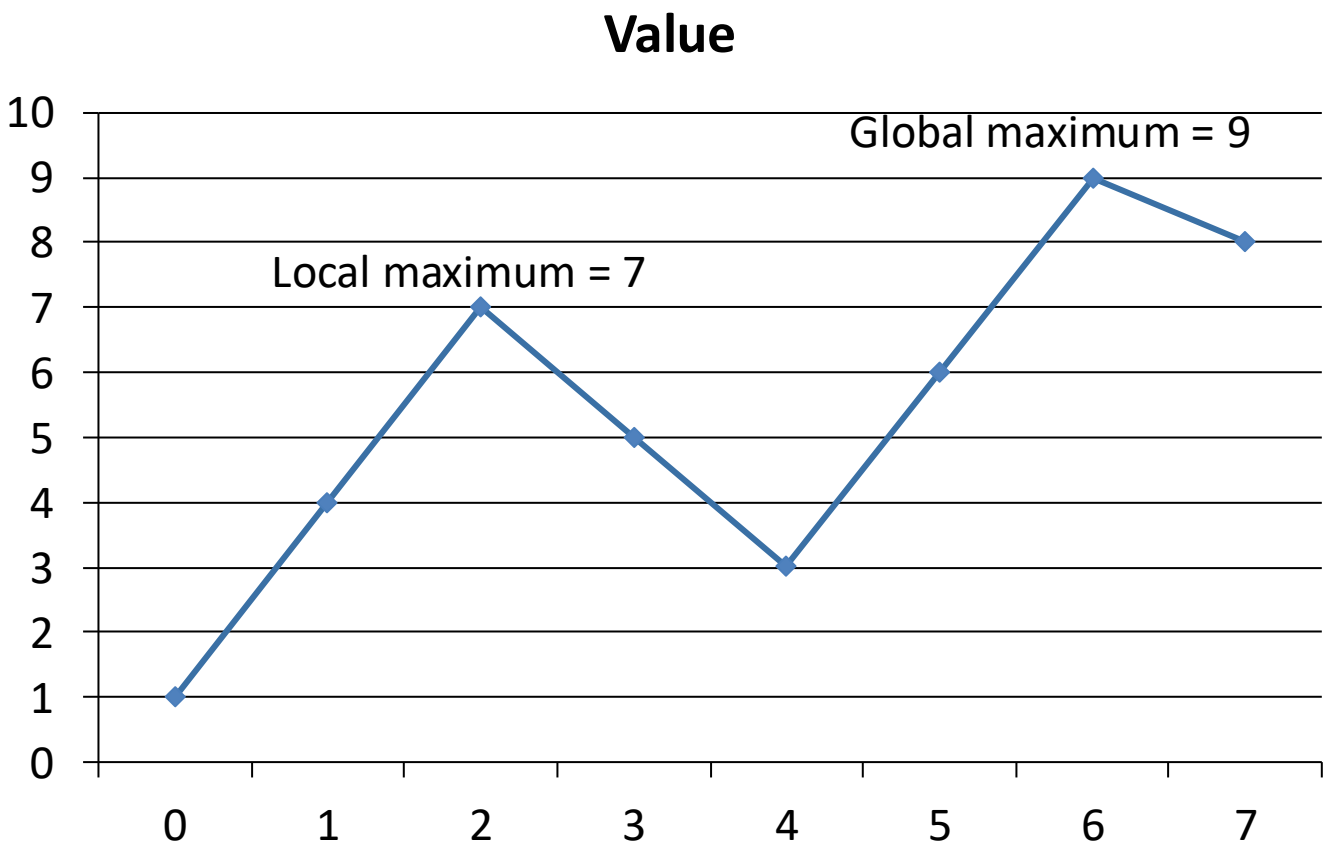


- 0 → 1 → 2 → 3
- Values: 1 → 3 → 5 → 8
- At index 3, both neighbors are worse.
- Result: final index = 3, final value = 8.

Rule: move only if a neighbor is better.

3. The Problem: Local Maximum

values = [1, 4, 7, 5, 3, 6, 9, 8]



- Start 0 → 1 → 2, then stop at value 7.
- 7 is a local maximum: both immediate neighbors are worse.
- But value 9 exists at index 6 — the global maximum.
- Hill Climbing cannot move 7 → 5 because that move is worse.

4. Random Restart Hill Climbing

Repeat Hill Climbing from several starting states and keep the best result.



- Advantage: less dependence on one initial state.
- Each run still uses ordinary Hill Climbing.
- Best result among all runs is returned.

5. Simulated Annealing

Sometimes accept a worse move to escape a local maximum.



- Hill Climbing would stop at 7.
- Simulated Annealing may temporarily accept 7 → 5.
- At high temperature T, worse moves are more likely.
- As T decreases, the algorithm becomes more selective.

6. What You Will Do in Practice 3

1. Trace Hill Climbing manually
2. Run Hill Climbing from different starts
3. Demonstrate a local maximum
4. Run Random Restart three times
5. Run Simulated Annealing five times
6. Compare the three methods

Submission: one Google Colab notebook with code, outputs, tables and short explanations.